

10. A student was given a sample known to be a mixture of magnesium oxide, MgO , and magnesium carbonate, MgCO_3 , and asked to determine the mass of magnesium carbonate in the mixture.

- (a) The first step was to accurately measure a mass of 1.5 g of the mixture into a crucible, using a balance.

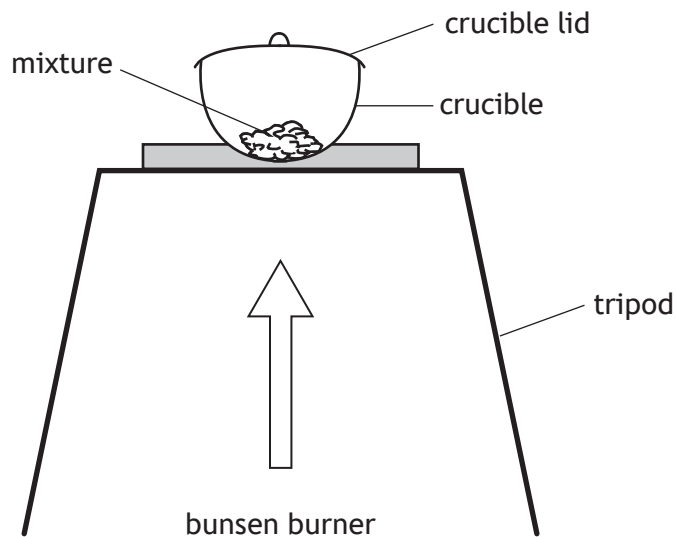
Describe **fully** how the student could carry out this step.

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- (b) Metal carbonates decompose on heating to give metal oxides and carbon dioxide gas.



The apparatus was used to strongly heat the mixture of MgCO_3 and MgO . The student used a pair of tongs to lift the lid from time to time.



- (i) Suggest why the student had to lift the lid from time to time.

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[Turn over



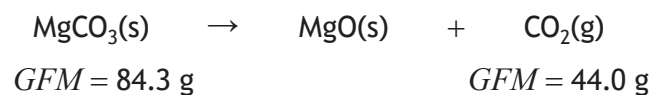
10. (b) (continued)

(ii) Explain why a Bunsen burner is a suitable method of heating this reaction.

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(c) Once the crucible had cooled, the final mass of the sample was determined. The mass of CO_2 lost is used to calculate the mass of MgCO_3 in the mixture. The results are shown below.

Mass of sample before heating	1.598 g
Mass of sample after heating	1.294 g



Calculate the mass, in g, of magnesium carbonate present in the mixture.

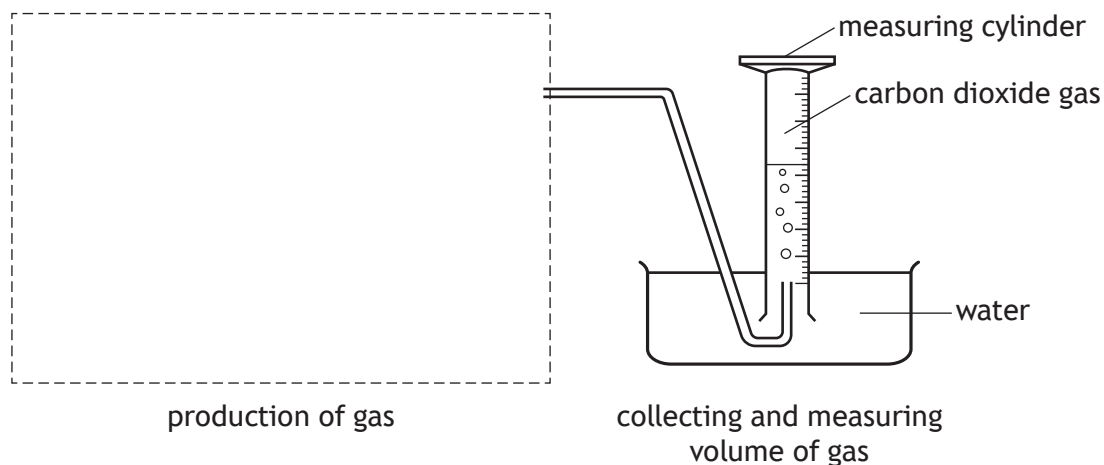
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10. (continued)

- (d) Another experiment to determine the mass of magnesium carbonate in the mixture involves measuring the volume of CO_2 produced.

This experiment is carried out by reacting the mixture of magnesium oxide and magnesium carbonate with excess acid. The volume of carbon dioxide gas produced can be measured by collecting the gas over water as shown.



- (i) Complete a labelled diagram to show an apparatus suitable for the production of gas.

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(An additional diagram, if required, can be found on *page 42*.)

- (ii) Suggest why carbon dioxide can be collected over water.

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[END OF QUESTION PAPER]

